Regression Splines

We consider a "ridgeless" setting slightly different than the slides

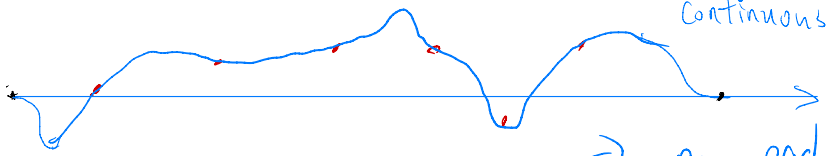
- Given $(x_1, Y_1) \cdots (Y_k, Y_k)$ $\quad x_1 < x_2 < \cdots < x_k$

spline function: f such that

$$f := \underset{g(x_i) = Y_i,\ i = 1, \cdots, k}{\operatorname{argmin}} \int \left(g''(x) \right)^2 dx$$

boundary condition

Continuous up to 2nd derivatives at knots



$\Rightarrow$ on each interval, f is a cubic polynomial.

- proof that f is piewise cubic, continuous up to second derivative:

Suppose h is arbitrary. $h(x_i) = h'(x_i) = h''(x_i) = 0 \quad \forall\, i = 1, \dots, k$

Consider perturbation $f \to f + \delta h, \quad \delta \in \mathbb{R}$

$$\frac{\partial}{\partial \delta} \int (f'' + \delta h'')^2 \, dx \Big|_{\delta = 0} = 2 \int f'' h'' \, dx \overset{\text{want}}{=\!=} 0 \quad \text{for all } h$$

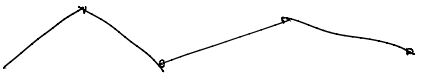
integration by parts $\int f'' h'' dx = -\int f''' h' dx$

$$= \int f'''' h \, dx \qquad (\text{used } f''(\text{boundary}) = 0, \; h(\text{boundary}) = 0)$$

$\Rightarrow f''''(x) = 0$ if x is not a knot

A simpler version

- How about $f := \underset{g(x_i)=y_i}{\operatorname{argmin}} \int (g'(x))^2 dx \Rightarrow f$ is piecewise linear

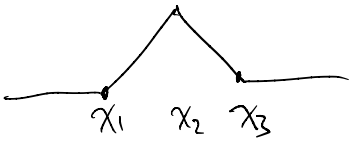


Reference for general splines:

Larry Wasserman "all of nonparametric statistics"

- Spine basis function

piecewise linear : given 3 consecutive knots

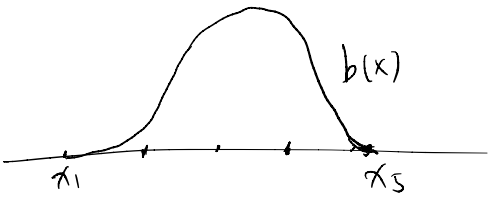


$b(x) \geq 0$, piecewise linear

$\int b(x)\,dx = 1$, supported on $[x_1, x_3]$

$b(x_1) = b(x_3) = 0$

piecewise cubic : given 5 consecutive knots



$b(x) \geq 0$, piewise cubic

$\int b\,dx = 1$, supported on $[x_1\ x_5]$

$b(x_1) = b'(x_1) = b''(x_1) = b(x_5) = b'(x_5) = b''(x_5) = 0$